National University of Computer and Emerging Sciences, Lahore Campus

Assignment 2

Course	Computer Organization and Assembly Language		
Course Code	EE2003	**Section**	BS (CY) – 3A
Semester	Fall 2026	**Page(s)**	2
Due Date	Sunday, 25 October 2026, 11:59 pm	**Total Marks**	50

Submission Format

- Upload the source code file (`.asm`) for each question, named `Q1.asm`, `Q2.asm` and `Q3.asm`.
- Upload a clear screenshot of the output of each question.

Question 1 **[10 Marks]**

Write an 8086-assembly language program that prints output by writing directly to video memory (segment `0xB800`). The program should first clear the screen, then display a 16-bit number in hexadecimal at a screen position of your choice. For example, the value `0x3A7F` should appear on screen as `3A7F`.

DOS and BIOS interrupts may not be used for printing; all output must go through display memory.

Question 2 **[15 Marks]**

Write an 8086-assembly language program that searches for a substring within a string. Use the paragraph below as the main string.

The program should take a substring (user input or defined in the data segment), display whether it was found, and if found, display the position of its first occurrence. Test it with at least one substring that exists in the paragraph and one that does not.

> *A sleepy cat named Byte lived inside an old computer lab. Every night, Byte chased the blinking cursor across the screen and tried to catch it with one paw. One evening, the cat found a plate of burgers next to the keyboard and ate every single one before the lab assistant returned. The assistant searched the desks, the registers, and even the stack, but the burgers were never found. Since that day, every program in the lab ends with a warning: never leave snacks near Byte.*

Question 3 **[25 Marks]**

Write an 8086 assembly language subroutine named `Last_Cap_String` that converts an ASCII string so that the **last** letter of each word is uppercase and all other letters are lowercase. The address of the string must be passed to the subroutine **through the stack**, and the subroutine must preserve all registers it uses.

The string is null-terminated and may contain letters, digits, spaces and punctuation. A word is a contiguous sequence of alphabetic characters; any non-alphabetic character ends a word and must be left unchanged.

Example:

Input: `the CAT ate the Burger, 2 FRIES!!`

Output: `thE caT atE thE burgeR, 2 frieS!!`

Write a separate subroutine named `Print_Centre`, also taking its parameters through the stack, that uses video memory to:

1. Display the original string on the first row of the screen.
2. Display the converted string at the centre of the screen, with each capitalized letter shown in a **different colour** from the rest of the string.
3. Display the total number of words in the string on the row below it.